

Total No. of Questions : 8]

SEAT No. :

PD-4490

[Total No. of Pages : 2

[6403]-298

T.E. (Computer Engineering) (Honors)
STATISTICS AND MACHINE LEARNING
(2019 Pattern) (Semester - VI) (310503)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, necessary.

- Q1)** a) Explain Matrix and Vector algebra with example. [6]
b) What is Matrix factorization concept explain with example? [6]
c) Explain systems of linear equations using matrices with one example. [6]

OR

- Q2)** a) What is concept of function and derivative, explain with example? [6]
b) What is concept of Multivariate calculus? [6]
c) Write short note on chain rule, Jacobian and Hessian. [6]

- Q3)** a) What is Machine Learning? What are the Applications of machine learning? [6]
b) How to design Learning system? [6]
c) What are the Machine Learning types? [6]

OR

- Q4)** a) What are the perspectives and issues in Machine Learning? [6]
b) Differentiate between supervised and unsupervised learning. [6]
c) Write short note on Reinforcement learning. [6]

P.T.O.

- Q5)** a) What are types of Regression? [6]
b) Write short note on simple regression. [6]
c) Describe Multivariable regression. [5]

OR

- Q6)** a) Differentiate between simple regression and multivariable regression. [6]
b) Describe in detail training and model evaluation of simple regression. [6]
c) Explain training and model evaluation of multivariable regression. [5]

- Q7)** a) Describe and explain the construction of decision tree. [6]
b) What are the classification of regression tree? [6]
c) Explain Bayes Theorem in detail. [5]

OR

- Q8)** a) Explain the working of Naïve's Bayes classifier. [6]
b) What are the types of Naïve's Bayes Model? [6]
c) Explain the advantages, disadvantages of Naïve's Bayes Model? [5]
